

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – AI Plagiarism Checker

Domain of Project – Education

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Problem Statement - AI-Driven Plagiarism Intelligence for Assignments

The Challenge: Academic integrity is the foundation of education, but institutions face increasing difficulty in detecting nuanced forms of academic misconduct, particularly when assignments are heavily paraphrased or generated by AI tools. Traditional plagiarism detectors operate on rigid string-matching algorithms and completely lack contextual sensitivity to an individual student's writing style or historical baseline. Without an automated, real-time way to evaluate individual linguistic footprints and track instructor feedback history, evaluators cannot reliably identify artificial text or structural anomalies in student submissions.

The Objective: Build an AI-powered Plagiarism Intelligence Agent that delivers context-aware academic integrity verification and deep stylometric insights for instructors, graders, and academic institutions. The solution should:

- Linguistic Persona Baseline – Evaluate and map a student's unique writing footprint based on vocabulary complexity, sentence structure distribution, and historical submission patterns.
- Contextual RAG Retrieval – Fetch and analyze a student's past assignments and corresponding instructor feedback from a local repository to establish an authentic baseline before evaluation.
- Multi-Modal Upload & OCR – Allow instructors to log and process assignments via digital text uploads or scanned handwritten/printed images, utilizing OCR for instant text extraction.
- Anomaly & AI Footprint Detection – Detect sudden vocabulary inflation, structural patterns typical of LLM generation, or major deviations from the student's historical grading profile.
- Multi-Agent System Architecture –
 - OCR Extraction Agent: Handles raw digital image processing and text serialization.
 - Contextual Retrieval Agent (RAG): Automatically queries the vector database for a specific student's historical writing baseline and grading logs.
 - Stylometry Comparison Agent: Cross-references the new text against the retrieved baseline to pinpoint sudden shifts in syntax, tone, and grammar habits.
 - Integrity Reporting Agent: Compiles the final analysis into a structured risk report, estimating an inconsistency probability percentage and grading verdict.

Proposed Solution

The AI-Powered Plagiarism Intelligence Platform is a multi-agent, context-aware verification system built using IBM watsonx.ai and Granite Models. Moving beyond rigid string-matching, it establishes an authentic "linguistic baseline" for each student to catch sophisticated AI text, paraphrasing and sudden shifts in writing style.

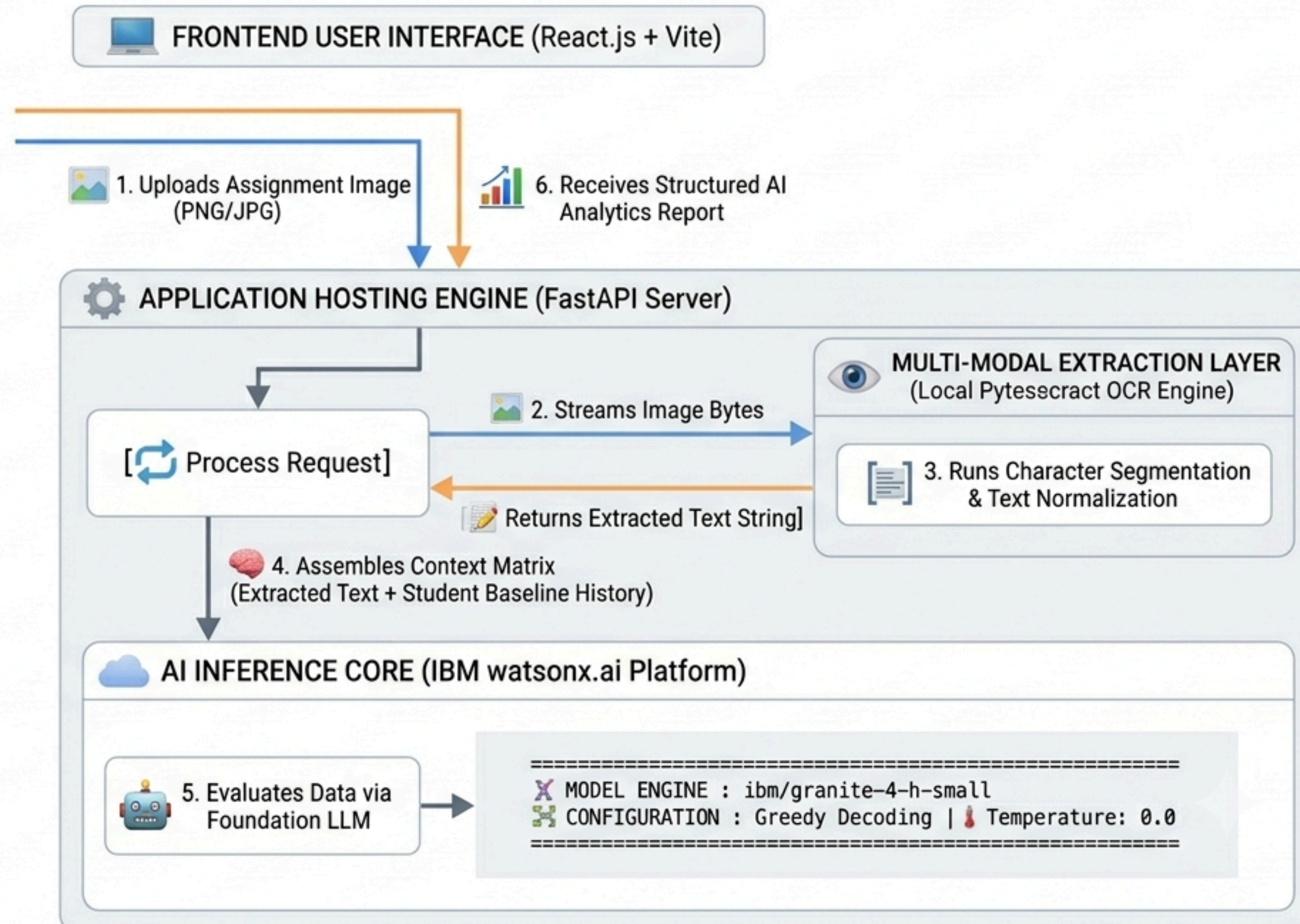
Key Architecture & Component Workflow:

- OCR Ingestion Agent: Extracts and serializes text from scanned handwritten or printed assignment images.
- Contextual Retrieval Agent (RAG): Queries ChromaDB using student metadata to fetch their unique historical submissions and past instructor feedback.
- Stylometry Comparison Agent: Employs IBM Granite (ibm/granite-13b-instruct-v2) via watsonx.ai to run side-by-side linguistic analysis, checking for unnatural vocabulary inflation or structural LLM patterns.
- Integrity Reporting Agent: Compiles findings into a precise risk percentage, flags anomalous text blocks, and outputs a final grading verdict (Consistent, Suspicious, or Highly Corrupted).
- Instructor Visualization Dashboard: Built using a React (TypeScript) + Vite frontend, the workspace features a centralized classroom risk matrix, dual-pane side-by-side style comparison screens, and automated integrity alerts to enable rapid, evidence-backed grading.

Technology Used

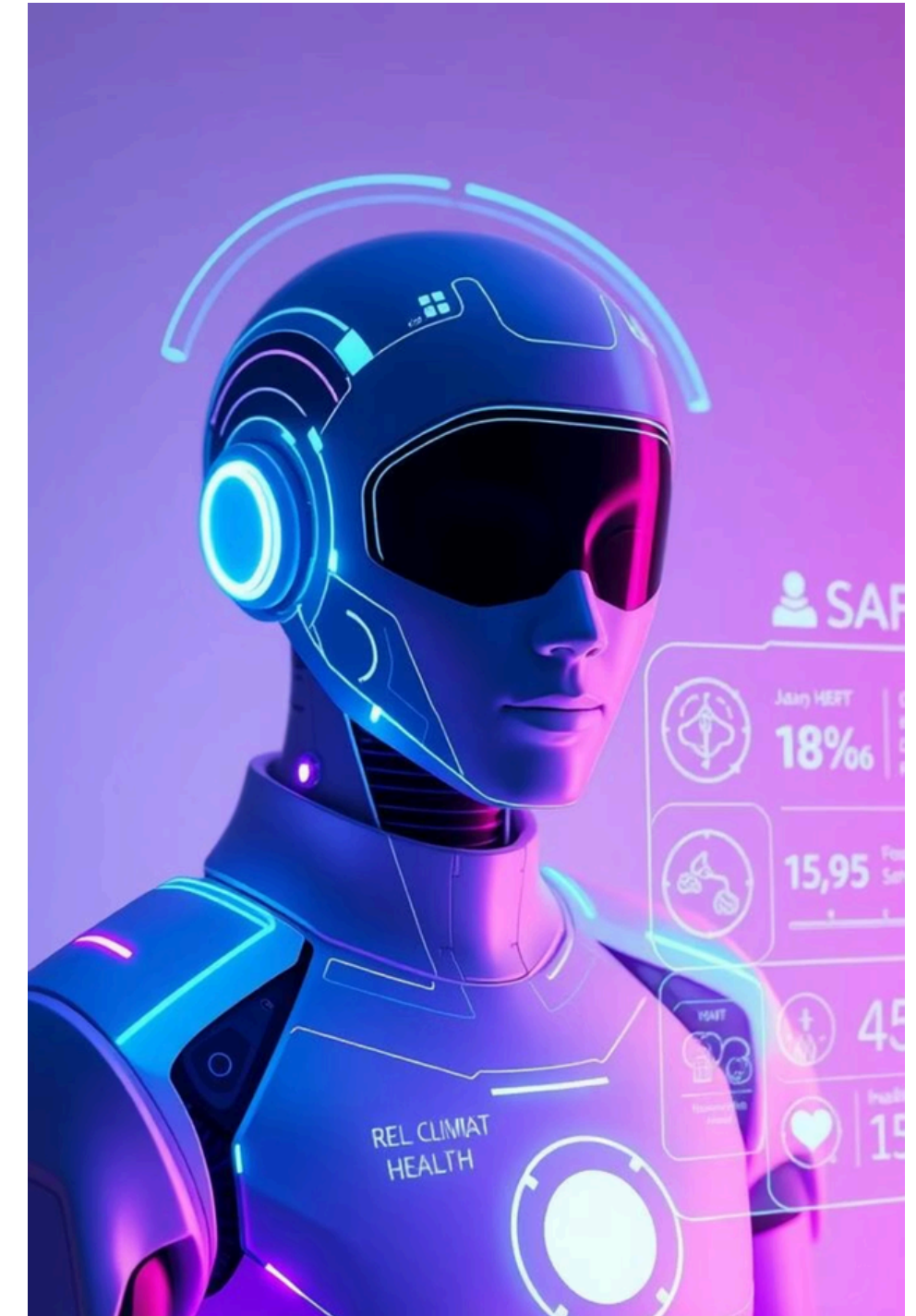
- **Framework:** FastAPI - Modern Python web framework for building APIs
- **OCR:** Tesseract - Extract text from assignment images
- **AI/ML:**
 - IBM Watsonx.ai - Model deployment and embeddings
 - IBM Granite Model - Natural language understanding and reasoning
 - RAG - Retrieve academic integrity references and generate analysis
 - Vector Database (FAISS/Chroma) - Store embeddings for efficient retrieval
- **Infrastructure:** IBM Cloud - Deployment and scaling
- **Data Storage:** JSON-backed store for student submission history

Architecture Diagram



Role of Agentic AI in the solution

Agentic AI transforms the platform from a reactive text-scanner into an autonomous, forensic auditing system powered by IBM watsonx.ai and IBM Granite models. Instead of running a basic keyword-matching script, it coordinates a collaborative multi-agent workflow where specialized digital agents handle distinct tasks—from local OCR character extraction and querying unique student writing baselines via RAG, to conducting side-by-side syntactic evaluations and drafting administrative risk profiles. This approach brings deep context awareness to academic integrity, allowing the system to analyze an individual student's historical vocabulary limits, structural habits, and past instructor feedback to flag sudden style anomalies rather than just static database matches. By enabling these agents to dynamically communicate and pass metadata back and forth, Agentic AI delivers a proactive, highly personalized, and evidence-backed digital auditor that detects sophisticated AI text and paraphrasing with precise clarity.

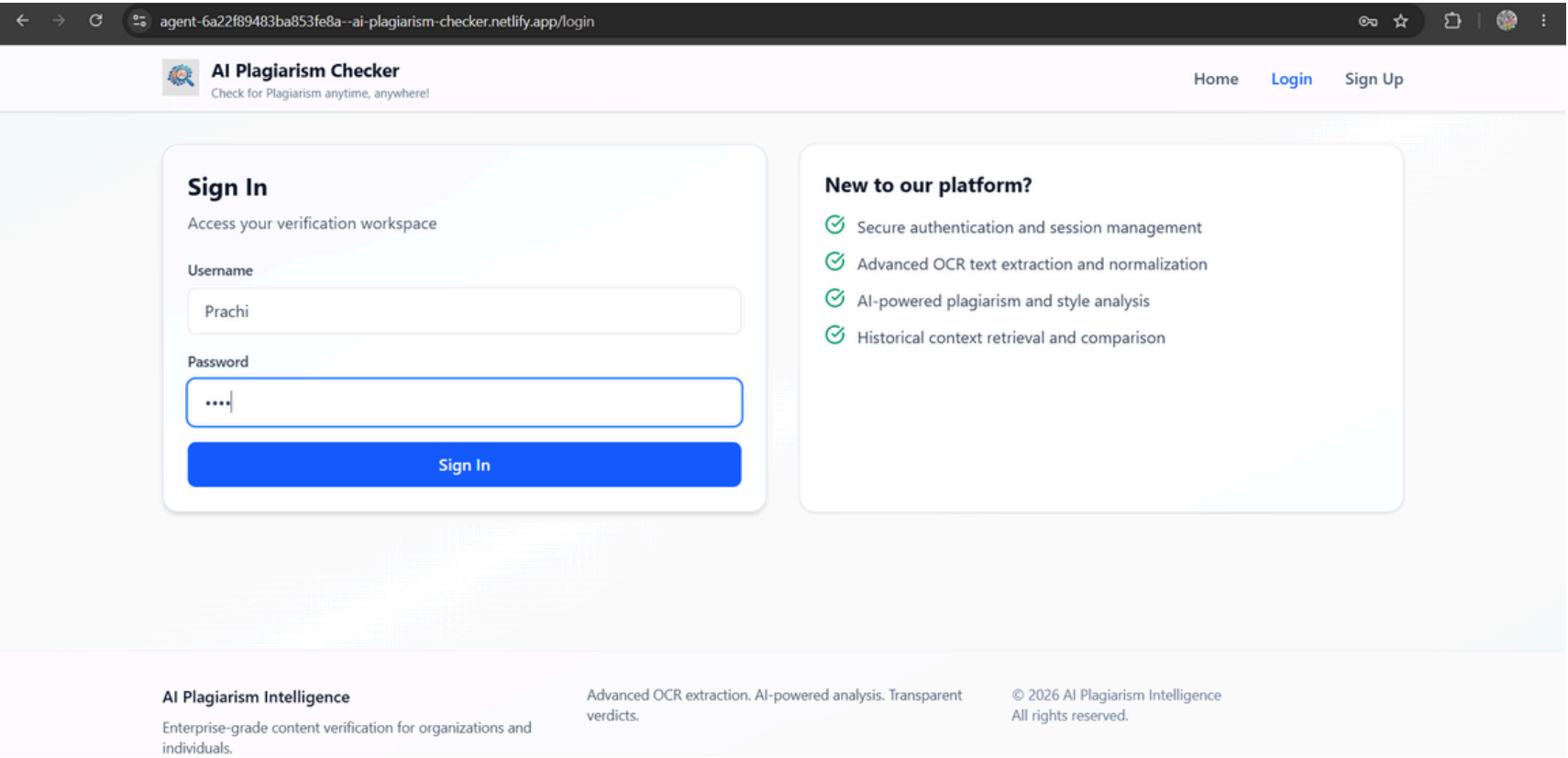
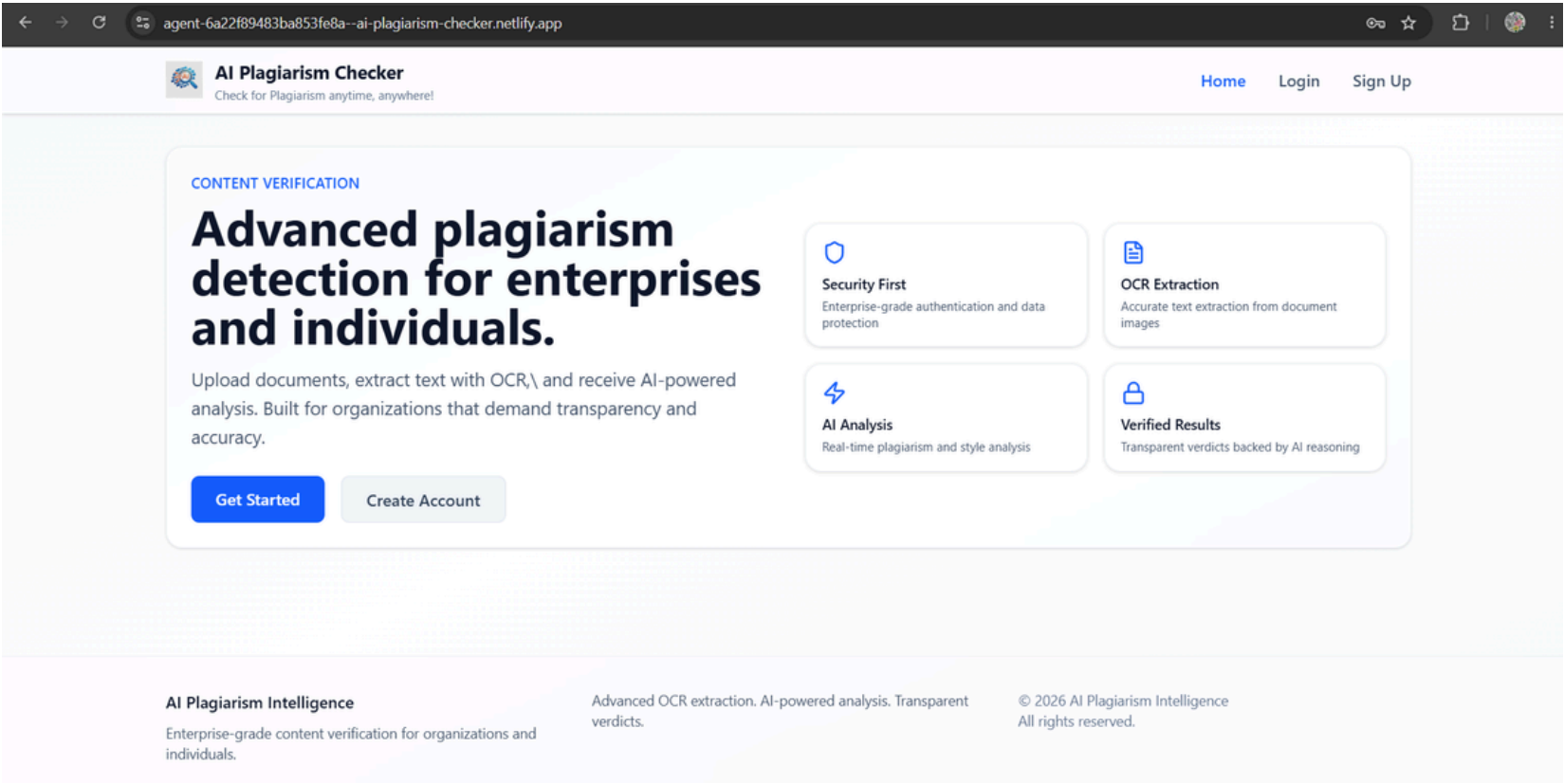


Novelty and Uniqueness

The proposed AI-Powered Plagiarism Intelligence Platform combines Agentic AI, Contextual RAG, and hyper-personalized stylometry into a single, comprehensive verification system.

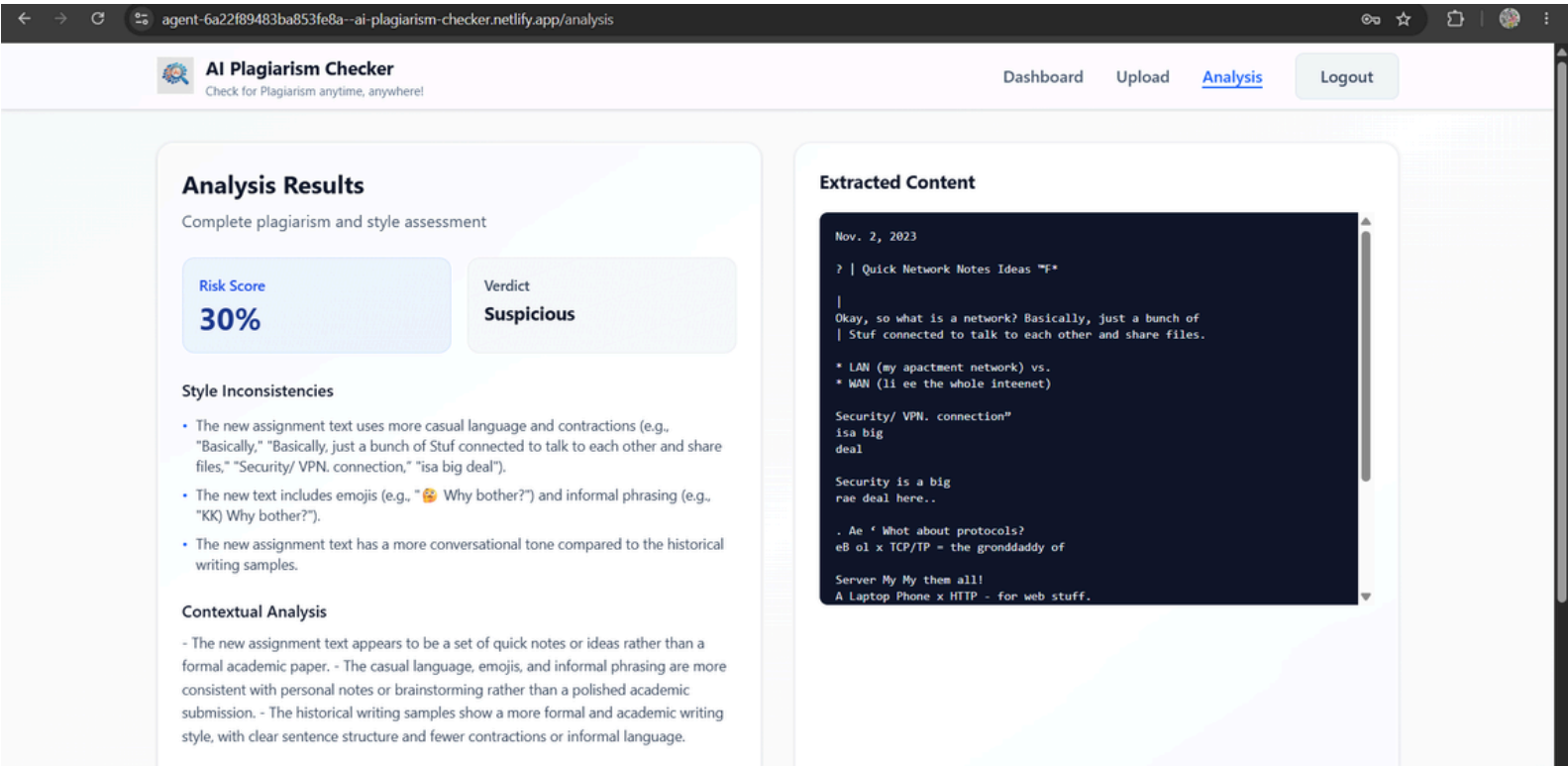
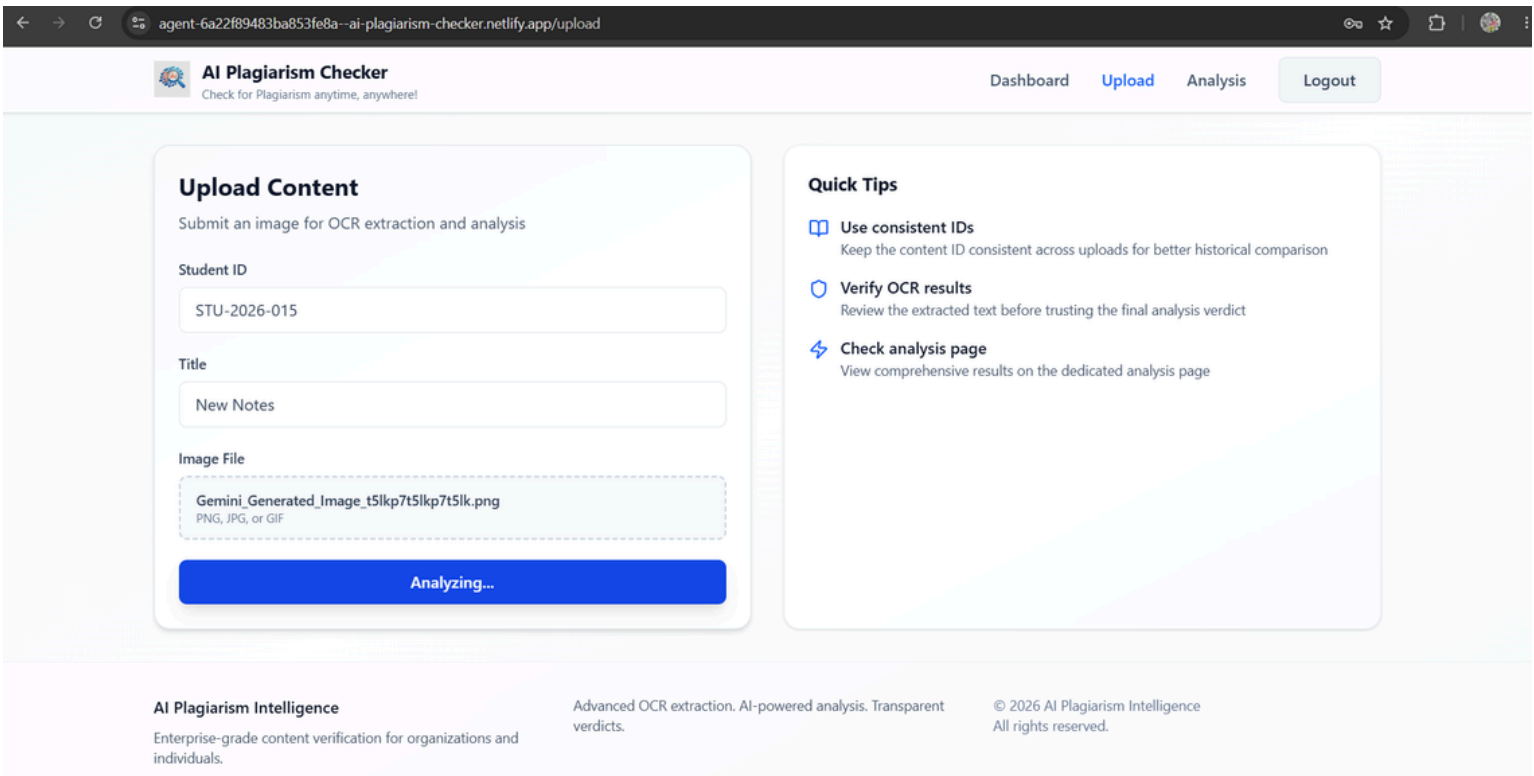
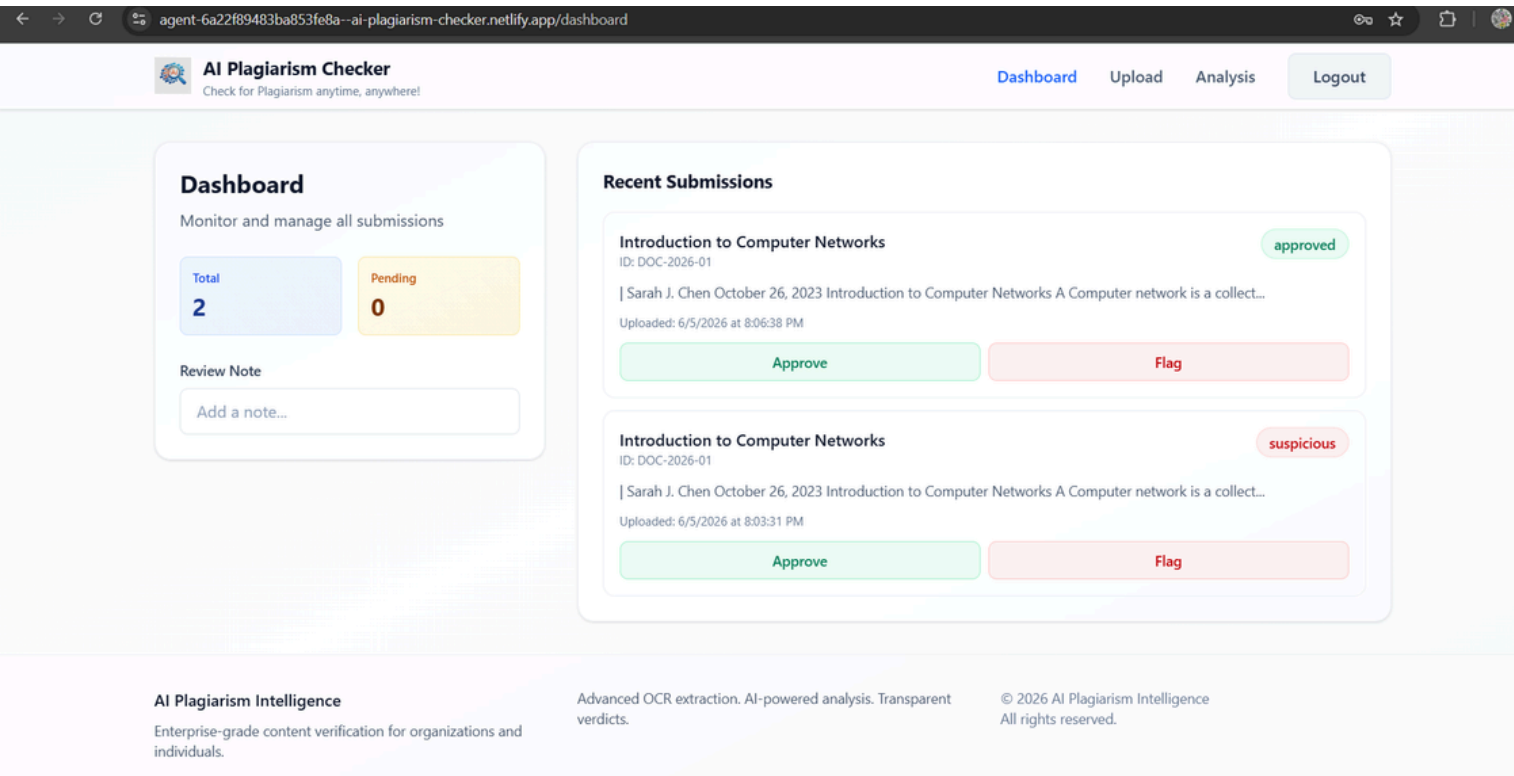
- **Multi-Agent Intelligence** – Multiple specialized digital agents collaborate autonomously to manage image processing, baseline memory retrieval, linguistic comparison, and administrative risk reporting.
- **Contextual RAG-Based Accuracy** – Integrates real-time, localized retrieval of a student's actual submission history with advanced AI reasoning, eliminating reliance on rigid external database matching.
- **Hyper-Personalized Footprints** – Evaluates new submissions against a student's unique writing baseline, accounting for their specific vocabulary limits and structural habit loops.
- **Multi-Modal Data Extraction** – Supports both digital file uploads and scanned handwritten or printed assignment images using a localized OCR pipeline.
- **Forensic Stylometry Focus** – Goes beyond simple copy-paste detection to expose sophisticated AI-generated text, extensive paraphrasing, and localized vocabulary inflation.
- **Enterprise-Ready Infrastructure** – Built securely on IBM watsonx.ai using IBM Granite models, ensuring strict governance, hallucination-free greedy decoding, and pipeline reliability.

Screenshots



Landing Page

Screenshots



User(Instructor) Dashboard

Links

Github Link – <https://github.com/Prashii06/ai-plagiarism-intelligence>

Deployment Link – <https://ai-plagiarism-checker.netlify.app>

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Future Scope

1. Integration with Campus LMS & Academic Ecosystems

The system can be directly integrated via webhooks into mainstream Learning Management Systems (LMS) such as Moodle, Canvas, and Google Classroom. This will enable the Plagiarism Intelligence Agent to automatically ingest student submissions the moment they are turned in, running background stylometric checks and updating the instructor's grading book interface seamlessly without requiring manual file uploads.

2. Advanced Intelligent Character Recognition (ICR) for Handwritten Papers

While the current system utilizes standard OCR for printed text, future updates will implement advanced Intelligent Character Recognition (ICR) machine learning models. This will allow the platform to accurately read, segment, and analyze varied student handwriting styles from physical exam papers or blue books, expanding its accessibility to offline, pen-and-paper academic settings.

3. Cross-Lingual Stylometry & Multilingual Anomaly Detection

Enhancing the system with multilingual support and cross-lingual translation mapping will prevent sophisticated translation-based plagiarism (where text is copied in one language, translated, and submitted in another). The agent will be able to evaluate structural writing footprints across multiple languages, ensuring academic integrity in diverse, bilingual, or regional linguistic environments.

Thank You!

Thank you for your time and interest.